

LLMs & GenAI in Robotics

CSCI 420-02 Robotics



WILLIAM & MARY

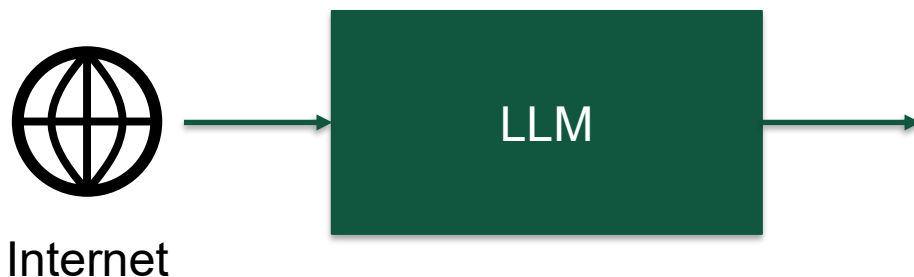
CHARTERED 1693

How do LLMs work?

- LLMs are prediction engines
- LLMs learn by example
 - Learn embeddings
 - Learn *associations*



What do LLMs learn?



What do LLMs learn?

Embeddings



Goal: represent words so

1. Similar concepts are **near** each other
2. The embedding is also a **vector** carrying its meaning

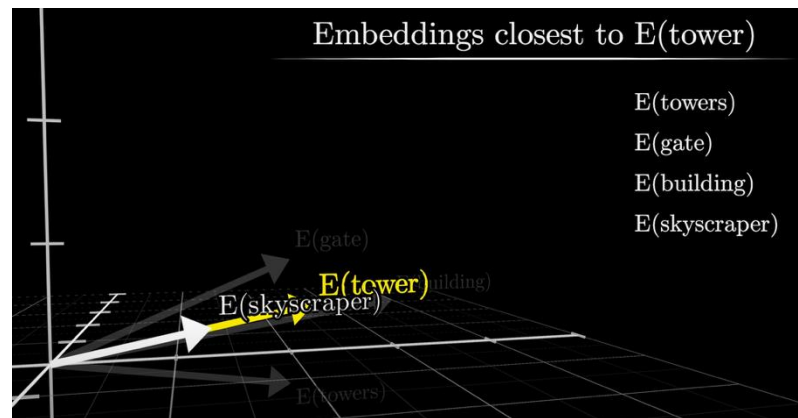
What do LLMs learn?

Embeddings



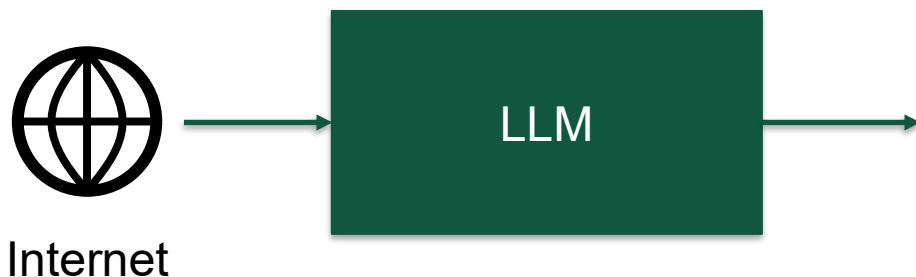
Internet

LLM

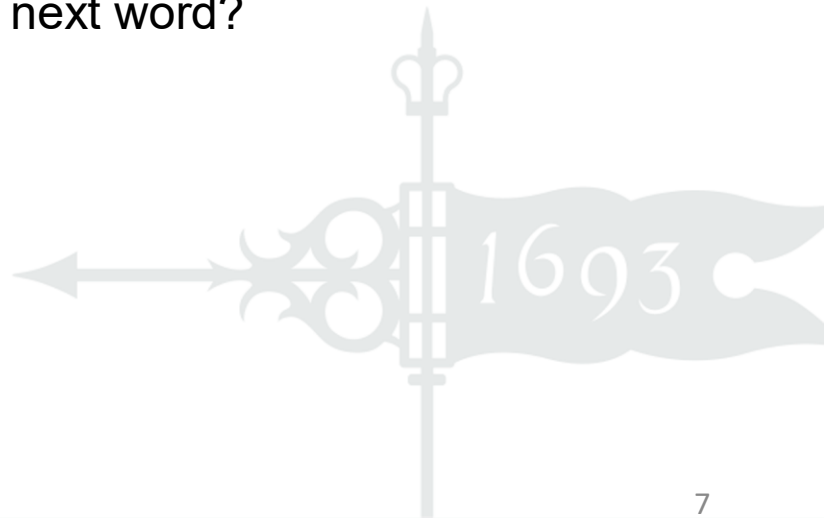


What do LLMs learn?

Associations



Goal: what is the most likely next word?



What do LLMs learn?

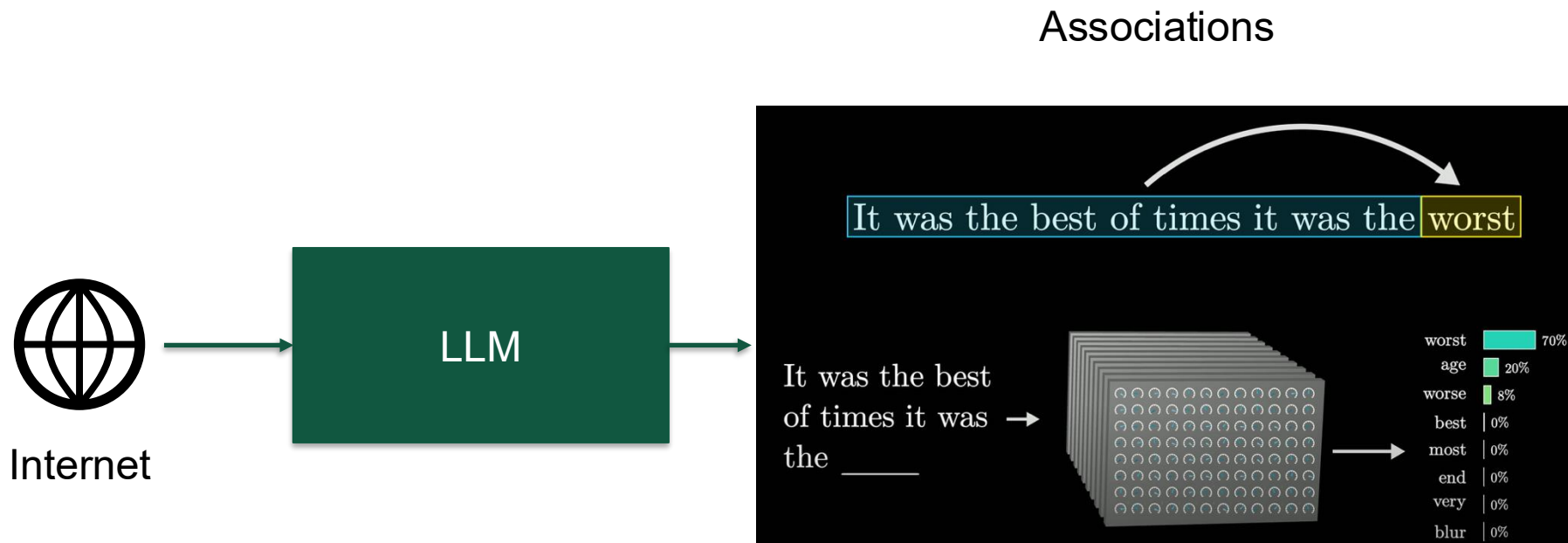
Associations



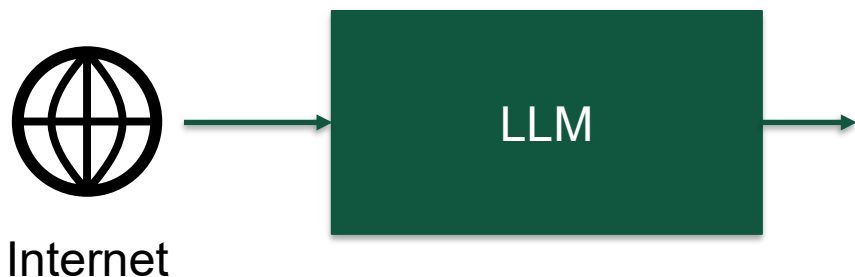
Goal: what is the most *likely* next word?

What defines likely?
Prior human examples!

What do LLMs learn?



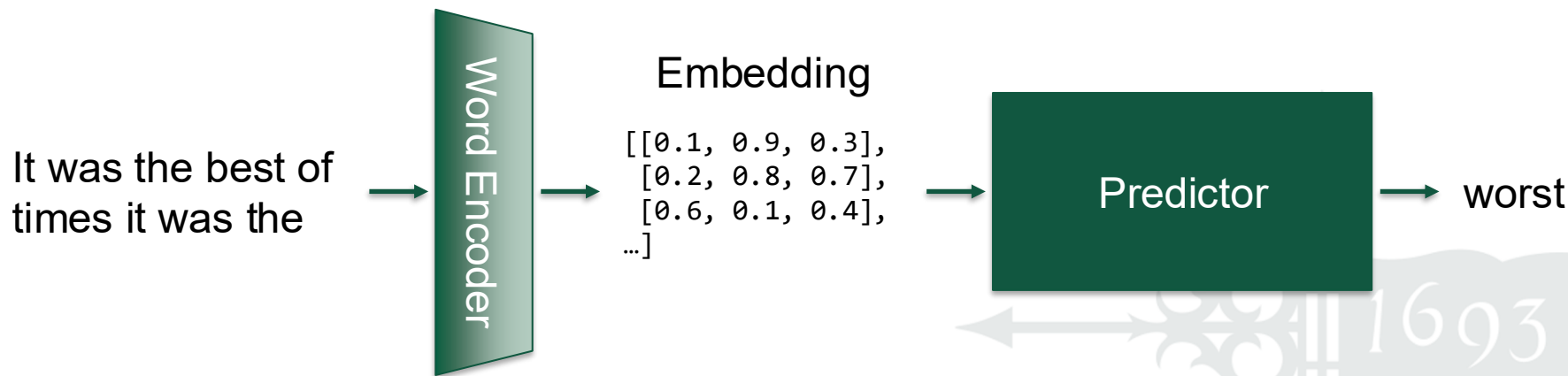
What do LLMs learn?



By training on prior human examples, LLMs learn:

1. Semantics of words relative to each other
2. To predict the next word from context

How do LLMs work?



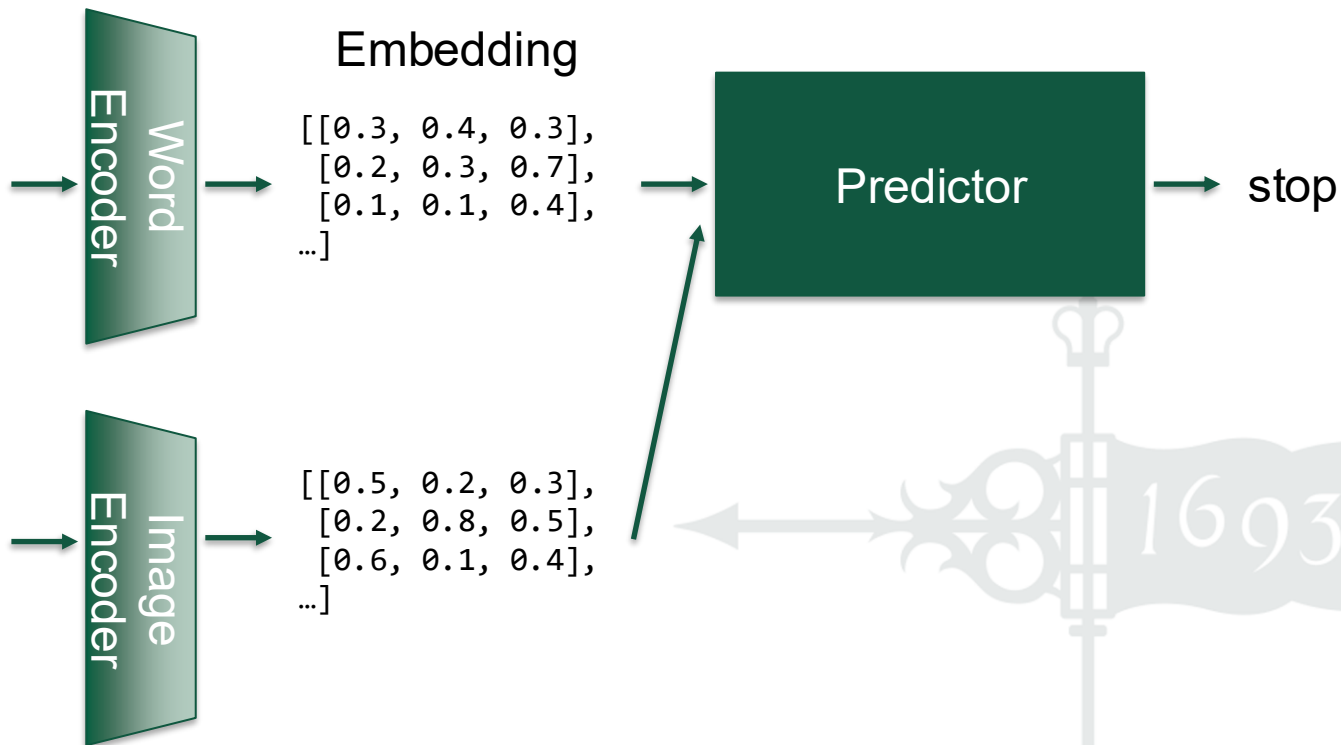
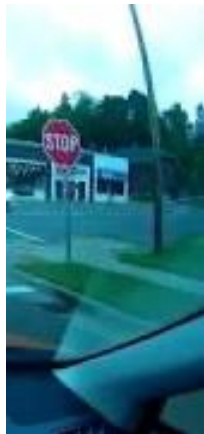
The Visual Variant

- Vision Language Models
 - Learn to interpret images by embedding



The Visual Variant

What sign
is this?



Leveraging LLMs for ADS

- Use LLMs to:
 - Interpret the world
 - Directly resolve requirement ambiguity
 - Formalize requirements



ODD-diLLMma

- ODD compliance checking using LLMs



ODD-diLLMma

- ODD compliance checking using LLMs
- Operational Design Domain
 - Defines environment conditions for ADS
 - Requirement for operation

ODDs are defined in natural language about the world

What is in ODD?

Company	DAS	ODD Semantic Dimension									
		Weather and Environmental Factors					Roadway Characteristics				
		Heavy Rain	Rain	Sleet	Ice	Bright Light	Low Light	Sharp Curves	On-Off Ramps	Intersections	Traffic Signals
comma.ai [9]	A	How do we identify these in the field?								×	
	A										×
GMC [18]	Super									×	×
	Traffic										
	Cruise										
	Aut										
T								×			
									✓	✓	✓

How close to an intersection counts?

Sharp curve?

How b

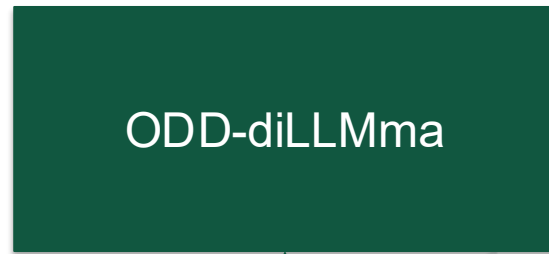
ODD-diLLMma

- ODD compliance checking using LLMs
 - LLMs can interpret the world
 - LLMs have seen previous human judgement to decide these ambiguities

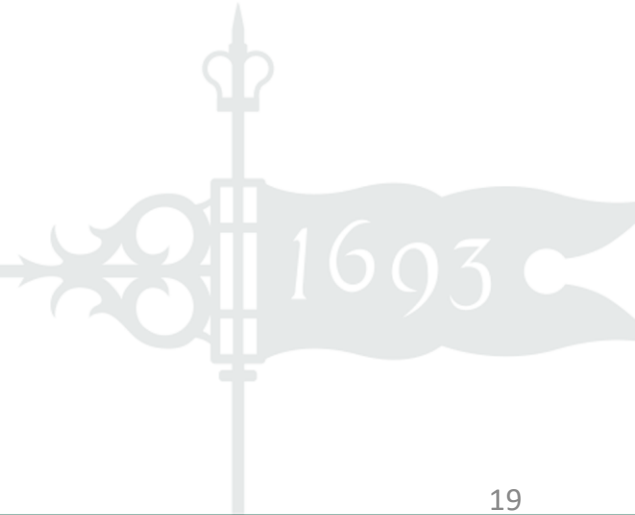
Use LLMs to check sensor data against the ODD!

ODD-diLLMma

- ODD compliance checking using LLMs



The ADS cannot
operate in heavy rain.



ODD-diLLMma

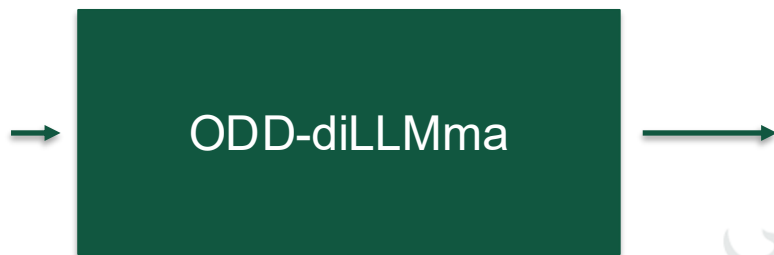
- ODD compliance checking using LLMs



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ODD-diLLMma

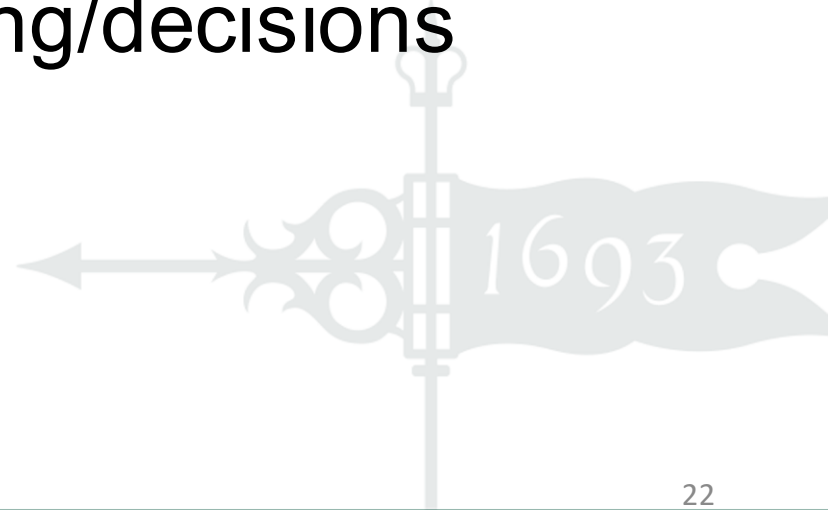
- ODD compliance checking using LLMs



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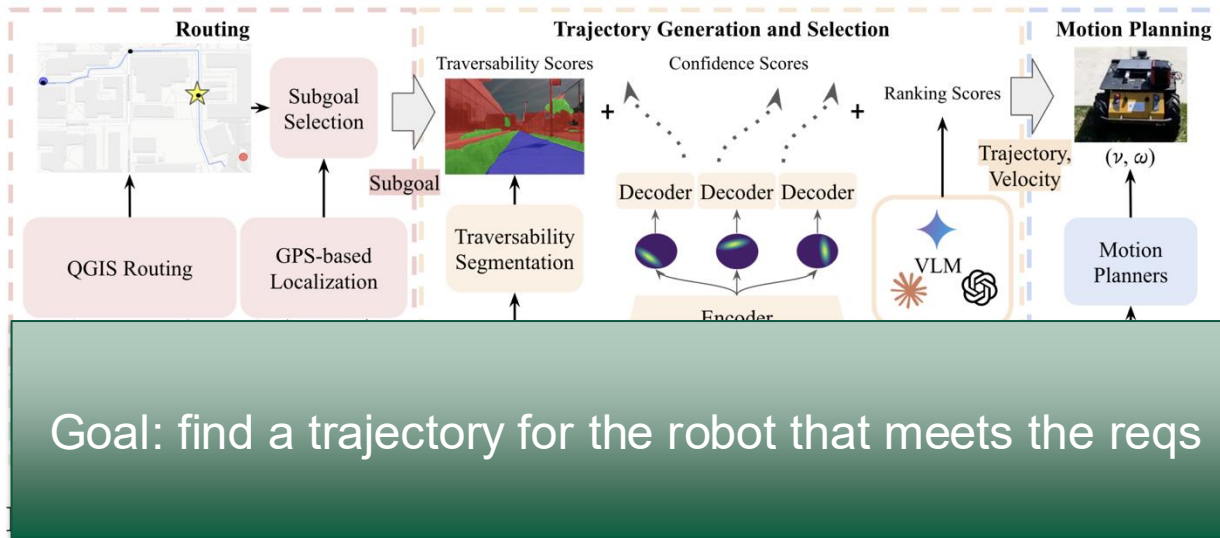
ODD-diLLMma

- Operationalize Natural Language Reqs.
- Over the real world through sensor data
- Mimicking human labeling/decisions



MOSU

- Multi-modal perception and On-road Scene Understanding for mobile robots



range waypoints, serving as high-level guidance for its trajectory generation system.

MOSU

- Multi
Scen
VLMs have observed this behavior – can they decide the requirement in practice?
ts
- Requirements:
 - Traversable terrain
 - Obeys traffic laws
 - *Obeys social cues*

Personal space is a cultural norm.

Robot must understand social cues and cultural norms for people to be comfortable around the robot.

The N trajectories are labeled with numbers $[0-N-1]$ from right to left in sequence. The goal is K meters at $[Right\ Front]$. Rank trajectories for social navigation.

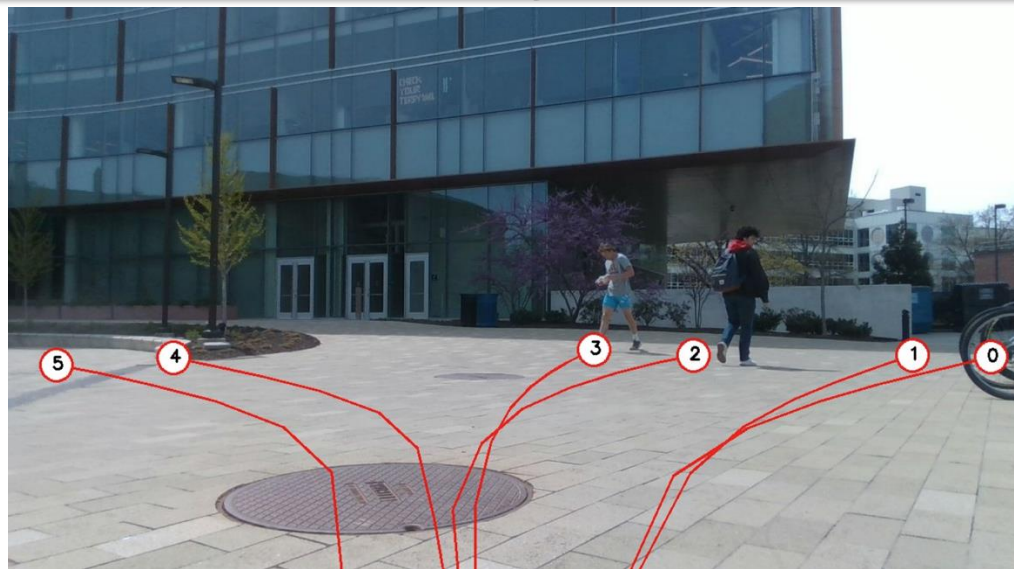
1. keep away from the groups of pedestrians. The robot has three mode, Normal, Slow, and Stop. If the people are approaching, the robot needs to Slow. If people are too close or there is no open space, the robots Stops.
2. follow the traffic rules, and if going across the street, the robot should keep in crosswalks.
3. recognize the traffic signs and behave accordingly.
4. avoid off-road terrain for small wheeled robots.

Given the picture, the target is at K meters $Front\ Left$. Rank the trajectories by the criteria. output the format: [robot mode], [ranked numbers], reason

MOSU

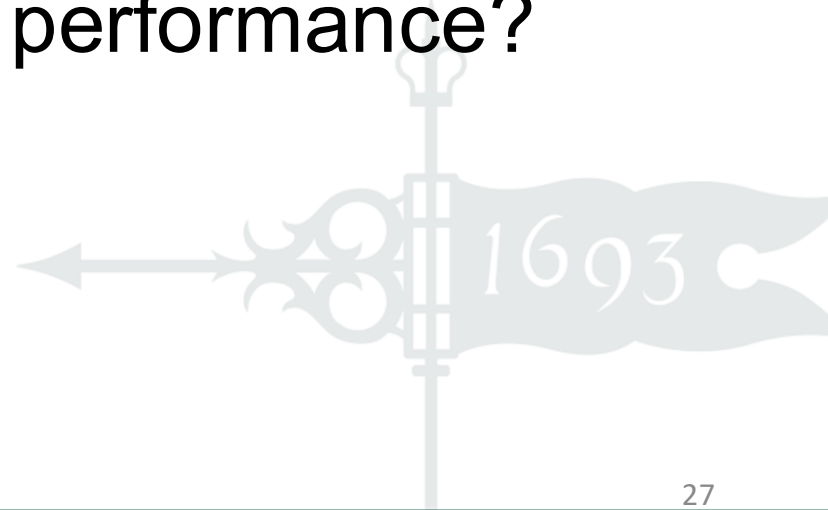
- Multi-Scene

How do we defend these rankings?



MOSU

- Operationalize “unwritten rules”
- How would we formalize these?
- What is the baseline for performance?



ScenicNL

- Translating real-world crashes into tests



ScenicNL

- Translating real-world crashes into tests
- Even for formalized requirements, testing is hard
 - Complex environment
 - Many states

How do we write down test cases anyway?

ScenicNL

- Translating

How do we write down test cases anyway?

- Scenic

- Probabilistic programming language
- Describes distributions of *scenarios*
- A test case samples from the distribution
- Ready-made integration with simulators

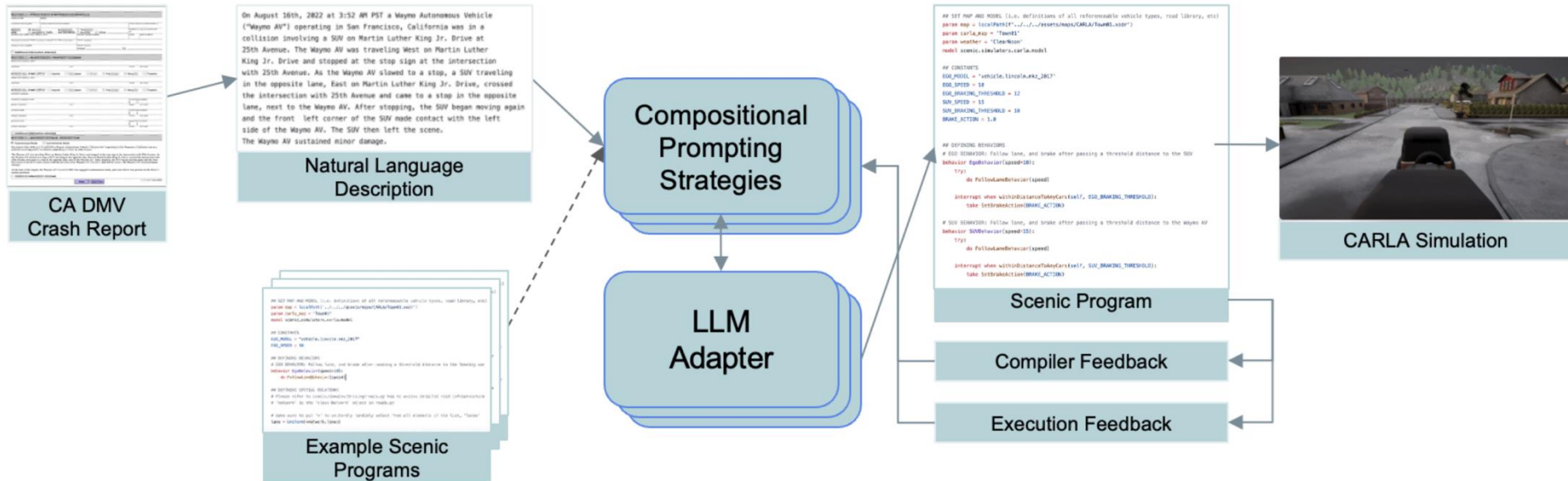
ScenicNL

- Translating real-world crashes into tests
- Humans have been driving for a long time
- Crashes represent difficult scenarios
- DMV has been documenting for years

How do we know what makes a good test case?

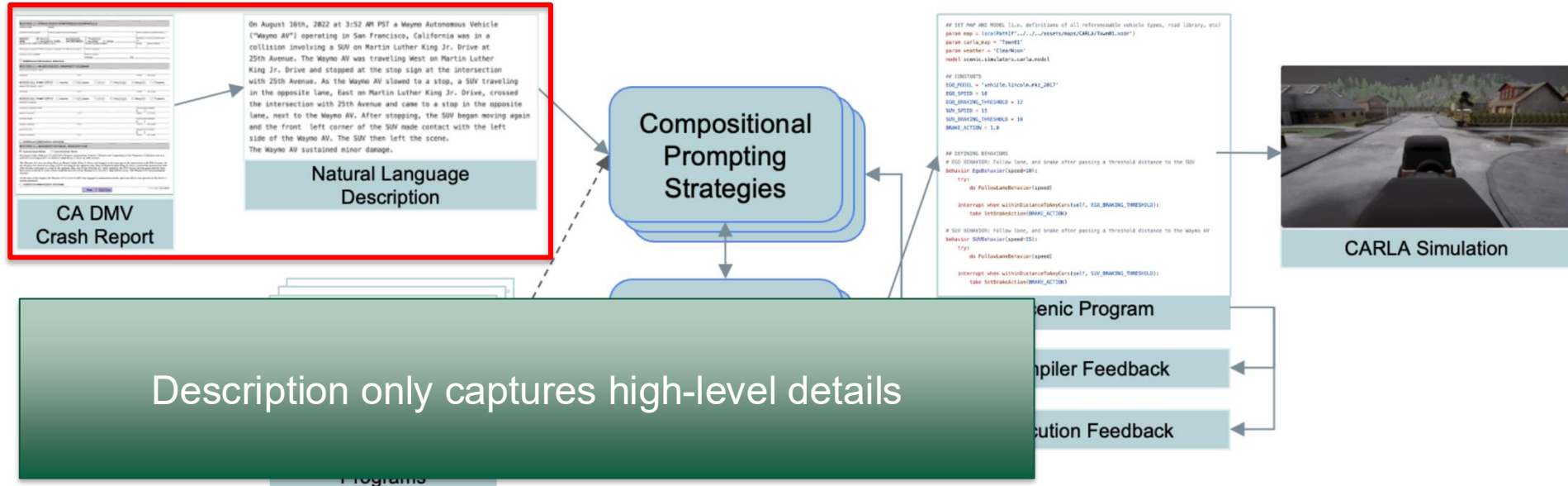
ScenicNL

- Translating real-world crashes into tests



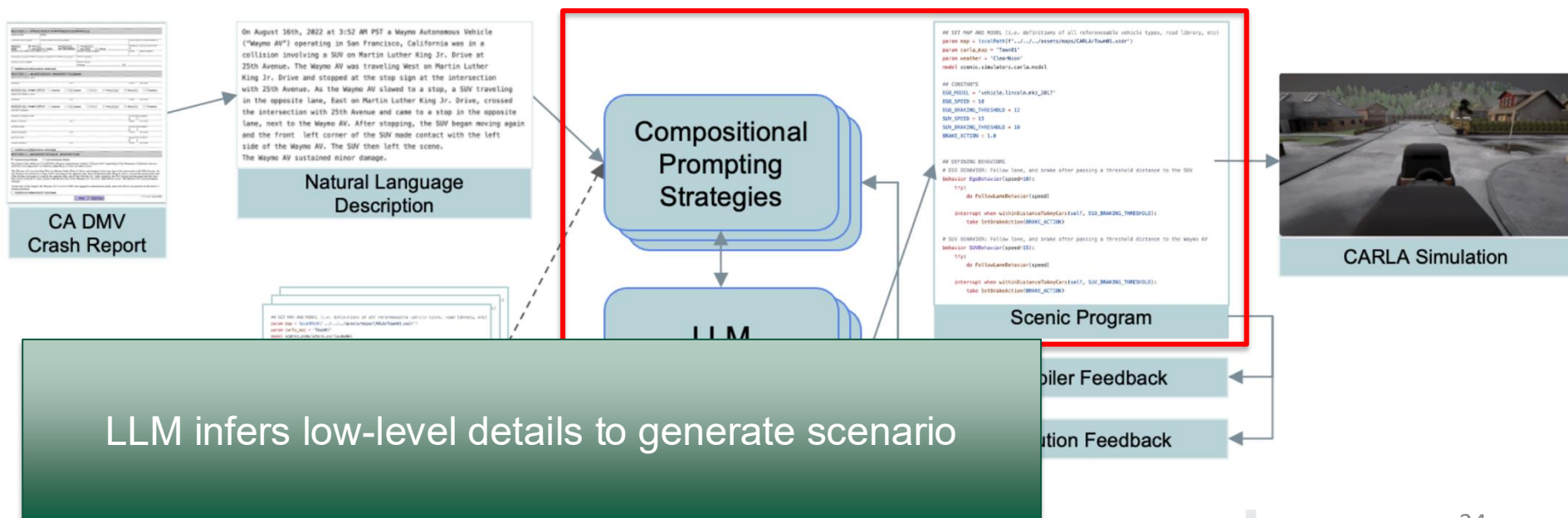
ScenicNL

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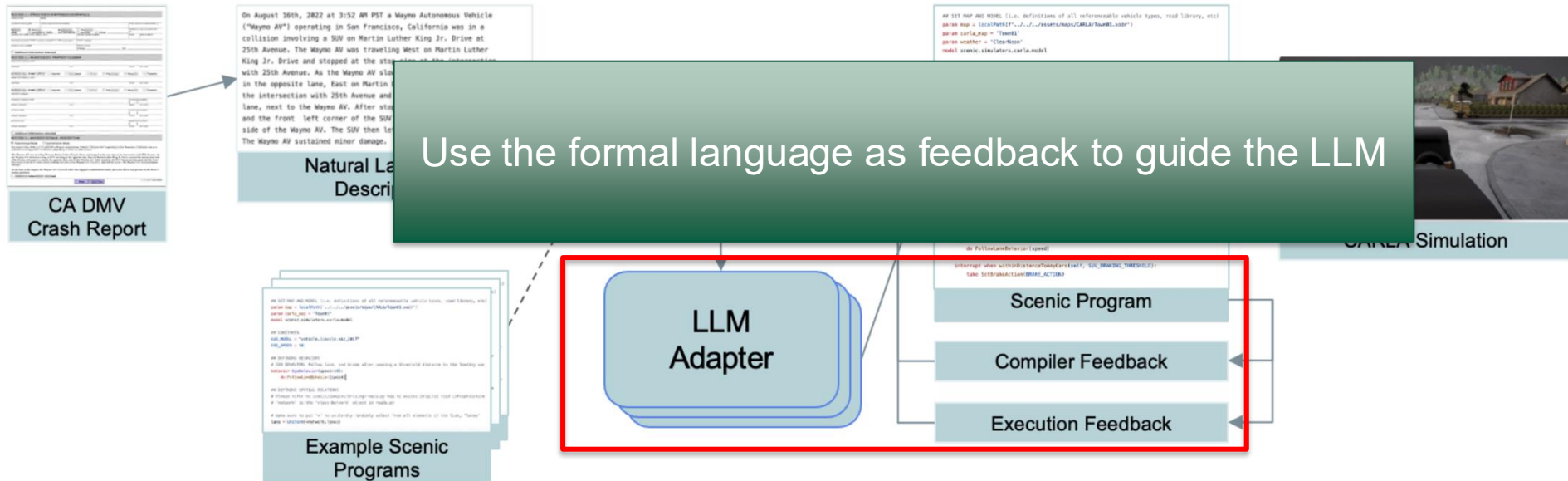
ScenicNL

- Translating real-world crashes into tests



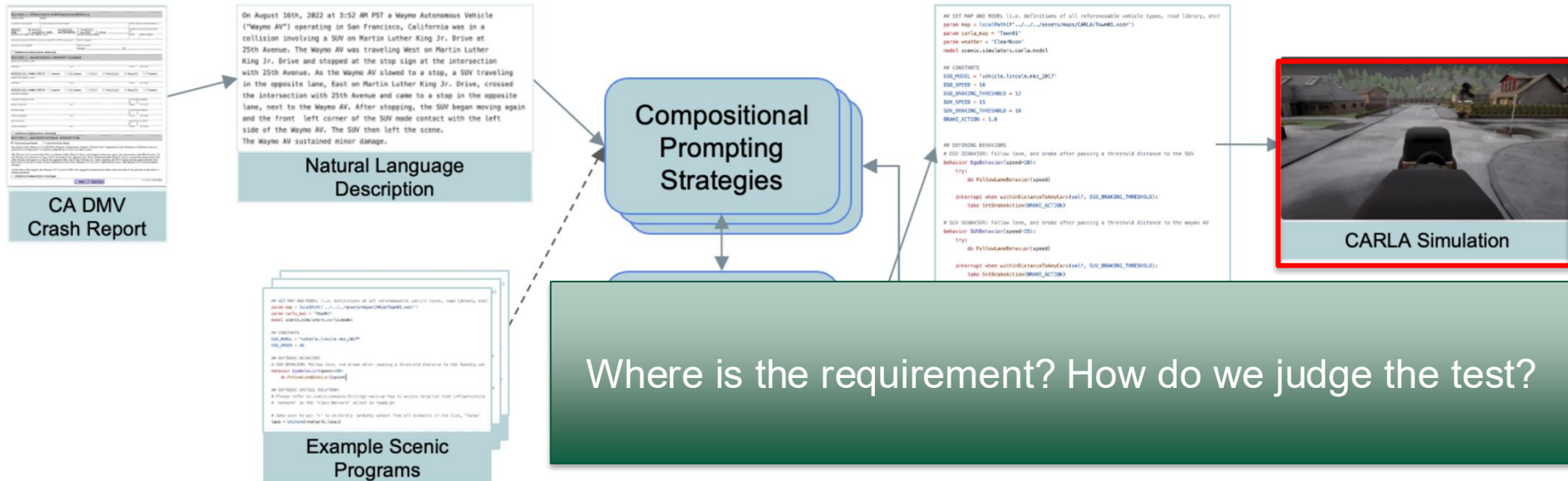
ScenicNL

- Translating real-world crashes into tests



ScenicNL

- Translating real-world crashes into tests



ScenicNL

- Leverage prior natural language
- Generate known-difficult test cases
- Automatically test ADS



Where can LLMs help?

- Interpret complex sensor data
- Directly decide ambiguous requirements
- Reason through informal/unwritten reqs
- Translate natural language to operation

